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Technical and Economic Feasibility of Hybrid Offshore Wind Systems in the Mumbai High Region: Three-Phase Implementation Framework

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Abstract

This study presents a comprehensive three-phase framework for integrating floating offshore wind systems with existing petroleum infrastructure in the Mumbai High to reduce fossil fuel dependence and support India's energy transition. The methodology combines a literature review, technical feasibility assessments, simulation modelling, and case study analyses of global projects, including Hywind Tampen and the North Sea Wind Power Hub.

The three-phase approach progresses from hybrid integration of 105 MW floating wind farms with existing gas turbines (Phase I), through smart interconnected power hubs using advanced control algorithms (Phase II), to HVDC subsea cable transmission for shore power export (Phase III). Simulation results confirm technical viability, recommending Star-F topology for optimal cost-reliability balance in smart power hubs.

Financial analysis demonstrates strong viability with \$558.4 M CAPEX, \$99.5/MWh LCOE, 18% IRR, and 9-year payback period. The project achieves 286,364 tonnes of annual CO₂ reduction and significant NO_x emissions savings. With 13% learning rate for floating wind costs, this research provides a replicable model for offshore renewable integration with oil and gas infrastructure, contributing valuable insights for India's offshore wind market development and global energy transition strategies.

Introduction

India's ambitious renewable energy targets—500 GW by 2030 and net-zero emissions by 2070—require rapid deployment beyond conventional onshore wind and solar resources. Offshore wind presents significant untapped potential along India's 11,100-kilometre coastline, with preliminary assessments indicating over 70 GW capacity in waters up to 200 meters deep. However, traditional offshore wind development models face economic and technical challenges in regions with moderate wind regimes and complex marine environments.

Mumbai High, India's largest offshore oil and gas producing region, offers a unique opportunity to address these challenges through innovative infrastructure integration. Operated primarily by ONGC, the basin hosts over 20+ offshore platforms that are approaching end-of-life as hydrocarbon reserves decline.

Rather than pursuing costly decommissioning processes estimated at \$12-18 million per platform, emerging global practices suggest repurposing existing infrastructure for offshore renewable energy development. This approach can reduce capital requirements by 32-45% while extending asset lifecycles and supporting India's energy transition objectives.

The fundamental challenge addressed by this research is whether hybrid offshore wind systems can be technically and economically viable in moderate wind regimes (6.5-8.0 m/s) when integrated with existing oil and gas infrastructure. Mumbai High's established marine logistics, operational expertise, and immediate energy demand from offshore platforms create conditions that may overcome traditional barriers to offshore wind development in emerging markets.

This study presents a comprehensive techno-economic analysis of implementing a 105 MW hybrid offshore wind system in Mumbai High, demonstrating technical feasibility, economic viability, and strategic value for India's offshore wind market development. The research establishes a replicable model for similar offshore oil and gas regions worldwide, positioning infrastructure reuse as a catalyst for accelerated renewable energy deployment and sustainable energy transition.

Literature Review and Global Context

Current Status of Offshore Wind Deployments Worldwide

The global offshore wind market exhibits a robust growth trajectory, despite occasional fluctuations in annual installations. According to the Global Wind Energy Council's 2025 report, 8 GW of new offshore wind capacity was grid-connected worldwide in 2024, bringing total global offshore wind capacity to 83.2 GW by year-end. While new additions were 26% lower than the previous year, 2024 still ranks as the fourth-highest year in offshore wind history, indicating the market's underlying strength and resilience. [1]

The offshore wind market has experienced remarkable expansion over the past decade, growing from 1.6 GW in 2013 to 8.0 GW in 2024, with its share of global new wind installations increasing from 4% to 7%. This growth trajectory reflects the improvement in technology maturity, cost competitiveness, and supportive policy frameworks across major markets. The compound annual growth rate (CAGR) of 4% demonstrates sustained momentum despite periodic market volatility. [1]

China maintains its position as the global offshore wind leader with 4 GW of new installations in 2024, representing 50% of the global market share with a cumulative capacity of 41.8 GW. The Chinese market's transition to grid parity—where offshore wind electricity receives the same remuneration as coal-fired power—demonstrates the technology's economic viability. China has successfully connected 5 GW in 2022 and 6.3 GW in 2023 without government support. 2024 saw a 36% reduction due to insufficient grid connections, complex maritime approvals, and slower-than-expected deep-water transition. Meanwhile, Europe commissioned 2.7 GW from nine wind farms across four markets, accounting for one-third of global capacity. The UK led Europe with 1.2 GW through key projects including Moray West (60 SGRE SG14-222 turbines) and Nearth na Gaoithe (28 SG8.0-167 turbines). Germany followed with 730 MW from Baltic Eagle (477 MW) and Gode Wind 3 (253 MW), while France added 658 MW through the Fécamp and Saint-Brieuc fixed-bottom projects plus the 25.2 MW Provence Grand Large floating wind project. [1]

Case Study: Hywind Tampen and floating wind integration

The Hywind Tampen project, located approximately 140 kilometres off the Norwegian coast in the North Sea, stands as a landmark initiative in the global advancement of hybrid offshore wind systems. It is the world's first floating offshore wind farm designed specifically to power oil and gas installations—the Gullfaks and Snorre fields operated by Equinor. The project merges floating wind energy with existing offshore petroleum infrastructure, establishing a viable blueprint for electrifying carbon-intensive offshore platforms. [2]

With a total system capacity of 94.6 MW, the Hywind Tampen wind farm comprises 11 Siemens Gamesa SWT-8.0-167 DD turbines, each with a nameplate capacity of 8.6 MW. Siemens Gamesa Renewable Energy is not only the turbine supplier but also serves as the operations and maintenance (O&M) contractor under a five-year service agreement [6][7]. All turbines are installed on floating substructures using spar-buoy foundations, designed to remain stable under harsh offshore conditions. These floating platforms are anchored at depths ranging from 260 to 300 meters, a technological feat made possible by Equinor's earlier experience. [6][4]

Hywind Tampen became operational in **the third quarter of 2022 and went into full operation as of August 2023** [4]. At full operation, it is expected to supply approximately **35% of the annual electricity demand** of the five Snorre A and B, and Gullfaks A, B and C platforms, displacing **about 200,000 tonnes of CO₂ and 1,000 tonnes of NO_x emissions** annually. [2]

From an engineering standpoint, Hywind Tampen exemplifies the seamless integration of floating wind substructures, dynamic subsea cabling, and advanced control systems. These components work together to coordinate the variable output of wind turbines with the steady requirements of gas turbine-based generation on offshore platforms. Electrical power is transmitted via 66 kV dynamic subsea inter-array cables, supported by two static export cables, each equipped with breakaway systems and specialised cable accessories. This energy is aggregated at an offshore hub and then distributed to the Snorre and Gullfaks platforms using the existing electrical infrastructure [8].

The project also incorporates digital twin models and predictive maintenance algorithms, improving operational efficiency and enabling real-time diagnostics—critical in the harsh offshore environment. These tools help optimise power balancing, turbine scheduling, and system resilience under fluctuating marine conditions. [9]

Hywind Tampen's success has set a technical and economic precedent for hybridising offshore platforms in deep-water regions like the Arabian Sea, where oil and gas infrastructure, such as that in Mumbai High, can benefit from similar floating wind integration models. Moreover, the project provides a testbed for scaling floating wind technologies, with insights into installation logistics, cost learning curves, and supply chain localisation that are highly relevant to emerging offshore wind markets globally.

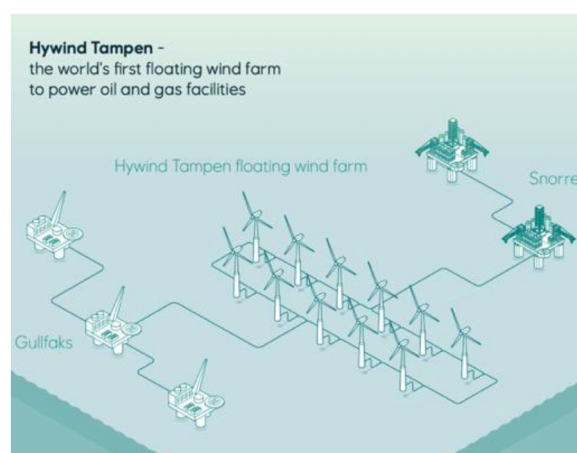


Figure 1—Illustration of Hywind Tampen wind farm. Source: Courtesy of Equinor ASA.

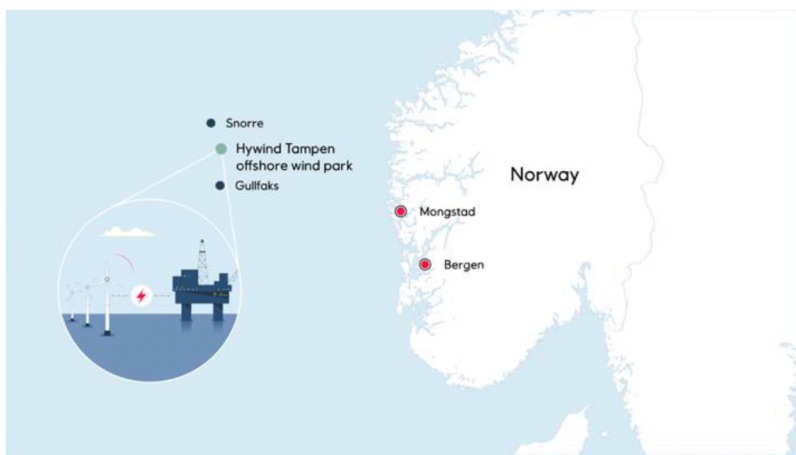


Figure 2—Illustration of Hywind Tampen wind farm. Source: Courtesy of Equinor ASA.

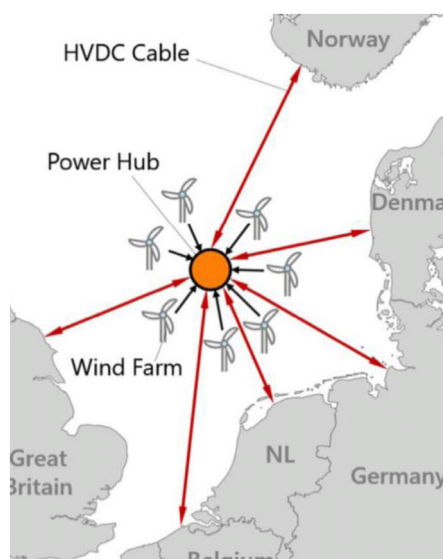


Figure 3—Illustration of the offshore wind power hub concept. Arrows indicate the short HVAC connections from wind farms to an artificial island, and longer HVDC links from this island to surrounding countries. [17]

Case Study: North Sea Wind Power Hub

The North Sea Wind Power Hub (NSWPH) represents one of Europe's most ambitious energy infrastructure projects, aiming to harness up to 180 GW of offshore wind energy potential by 2050 through an innovative hub-and-spoke model. Established in March 2017, the consortium brings together three leading transmission system operators—Energinet (Denmark), TenneT (Germany), and Gasunie (Netherlands)—working toward launching the first operational project by the early 2030s. [11][12][13]

The project's core concept involves constructing artificial energy islands strategically positioned in the North Sea to serve as collection points for large clusters of offshore wind farms. Rather than traditional radial connections, these modular hubs will link multiple countries through high-capacity transmission lines, creating an interconnected grid topology that enables redundancy, cross-border power-sharing, and load balancing among participating nations. From a technical perspective, the energy hubs support sophisticated multi-terminal HVDC infrastructure, hybrid interconnection capabilities, and Power-to-X technologies for offshore hydrogen production. This comprehensive approach enables flexible conversion of surplus wind power into storable, dispatchable fuels—critical for decarbonising heavy industry and maritime transport—while significantly enhancing grid resilience and cost-effectiveness. [14][15][16]

Regional Feasibility – Mumbai High Region

Wind Resource Suitability

The Mumbai High region, located off the western coast of India, shows moderate but promising potential for offshore wind development. Wind resource assessments indicate that average wind speeds in the broader Maharashtra coastal zone range between **6.5 to 8 m/s**, which is within the operational threshold for commercial offshore wind farms. [20] Also, according to the comprehensive study by Mani Murali et al. (*Site selection for offshore wind farms along the Indian coast*, 2014), one of the most viable offshore wind zones in India—termed **Feasible Region 1**—spans approximately **32,000 km²** in the northeast Arabian Sea, directly **off Mumbai and Ratnagiri**. This area features bathymetry between **20 m to 75 m**, making it compatible with fixed-bottom and floating offshore turbine foundations. [21]

Wind speed is the cornerstone determinant for offshore wind viability. Seasonal data derived from QuickSCAT climatology (1999–2009) indicates that **Region 1 experiences wind velocities ranging from 3.3 m/s to 10.2 m/s** across different seasons. Notably, **during the summer monsoon**, wind velocities can peak at **10.2 m/s**, offering the highest potential for power generation. Even in the winter and post-monsoon seasons, wind speeds in parts of Region 1 exceed the 5 m/s threshold necessary for offshore energy viability. [21]

The **numerically generated offshore wind maps by NIWE**, illustrated in Figure 2, further confirm that the western coastline consistently experiences wind speeds ranging from **6.5 to over 8.0 m/s** at hub heights of 100 meters. These values are within the design specifications of modern offshore wind turbines and comparable to wind conditions in early-phase North Sea projects.



Additionally, temporal wind consistency is crucial for energy yield stability. According to INCOIS/MoES wind occurrence maps (Figure 1), the Mumbai High corridor experiences **>6 m/s winds for 275+ days annually**, and **>8 m/s winds for 100–150 days/year**, especially during the southwest monsoon (June–September), which aligns well with India's seasonal peak demand periods.

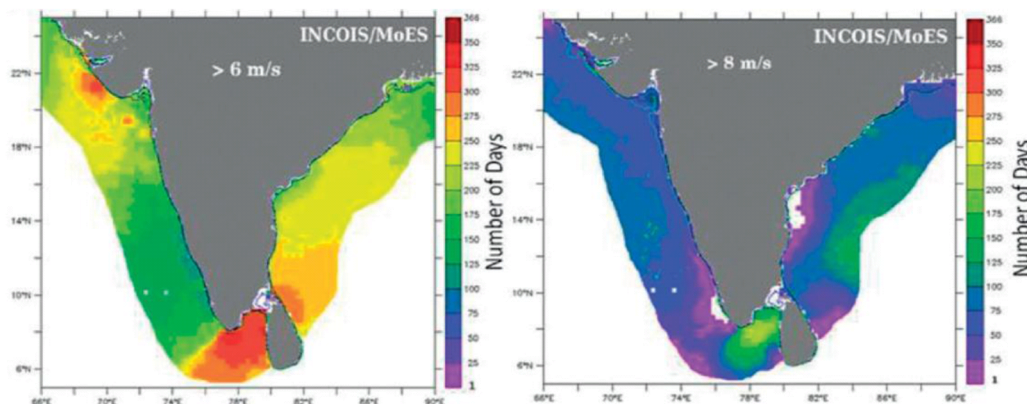


Figure 6—Offshore wind potential maps for the Indian coast by Earth System Science Organisation (ESSO)-INCOIS. [19]

Overcoming Low Wind Speed Challenges in Offshore Wind Development

Despite concerns that moderate wind regimes pose a constraint to offshore wind development in regions like Mumbai High, recent evidence suggests otherwise. A pivotal study by Iqbar et al. (2022), focusing on offshore wind deployment in Malaysian waters with average wind speeds of 6.8 m/s, revealed that capacity factors of 38% and a Levelized Cost of Energy (LCOE) of USD 165.52/MWh were achievable. This represented a 39% reduction compared to conventional generation, even in a low-wind setting. [23]

A key feature of the study was the use of a hybrid system architecture, combining offshore wind turbines with gas turbine backup—a configuration highly relevant for the Mumbai High cluster, where ONGC's offshore platforms already use gas-based power systems. The economic outcomes were particularly encouraging: payback periods of approximately 4 years for greenfield and 12 years for brownfield projects were recorded. Furthermore, positive Net Present Values (NPV) were maintained across all sensitivity scenarios, even under a 19% drop in energy output.[23]

The technical success in Malaysia was underpinned by intelligent turbine selection—specifically, 155 m rotor diameters and hub heights of 110 meters, which allowed efficient energy capture even in marginal wind environments. These design parameters are readily translatable to the Mumbai High context, especially with India's growing access to large-scale turbine models. [23]

Given Mumbai High's proximity to major coastal load centers, the potential for retrofitting ageing oil platforms, and access to existing marine logistics corridors, replicating or even exceeding Malaysia's performance benchmarks appears entirely feasible. When paired with hybrid power solutions and smart grid integration, this reinforces the technical and financial case for offshore wind deployment in India's western seaboard—even under modest wind conditions.

Reuse of Existing Oil & Gas Infrastructure

India's ageing offshore infrastructure in the Mumbai High basin presents a timely and economically attractive opportunity to support hybrid renewable energy development. Operated primarily by ONGC, the Mumbai High cluster hosts over **100 offshore platforms**, many of which are approaching the end of their hydrocarbon production lifecycle. Rather than pursuing full-scale decommissioning—a process both capital-intensive and environmentally disruptive—there is growing global momentum to **repurpose these structures for offshore wind integration**.

A comprehensive study by [Braga et al. \(2022\)](#), analysing the conversion of Brazilian offshore oil and gas infrastructures into renewable energy generation plants, revealed that wind power installations offer a compelling economic alternative to conventional decommissioning approaches. The research demonstrated that CapEx discount rates of 32-45% were achievable when offsetting traditional removal costs through wind power project development. [24]

A key feature of the Brazilian study was the strategic repurposing of fixed offshore platforms—particularly steel jacket, concrete base, and caisson structures—as foundations for 15 MW horizontal-axis wind turbines (HAWTs). This approach proved highly relevant for ageing offshore installations, where existing infrastructure can be retrofitted rather than completely removed. The economic outcomes were particularly encouraging: wind power project costs of USD 39 million compared favorably against decommissioning expenses ranging from USD 12.7-17.6 million per platform, representing significant capital savings while generating renewable energy revenue streams. [24]

From a technical standpoint, the Brazilian case emphasised intelligent **site selection and structural reuse**—focusing on regions with **wind speeds >6.5 m/s** and water depths between **6–30 meters**. These criteria are remarkably similar to conditions found in the **Mumbai High region**, where several platforms are located in **shallow to moderate depths (35–75 m)** and experience **annual wind speeds in the 6.5–7.0 m/s range**, based on NIWE and INCOIS datasets.

Given the global challenge of offshore platform decommissioning, the potential for extending asset lifecycles through renewable energy conversion, and access to established marine logistics networks, replicating Brazil's innovative approach appears entirely feasible for other offshore oil and gas regions worldwide. When combined with the environmental benefits of delayed decommissioning—including reduced carbon emissions and preserved marine ecosystems—this reinforces both the technical and financial case for infrastructure repurposing as a sustainable alternative to traditional field abandonment practices.

Technical Design of the Hybrid Offshore System for Mumbai High

TechnologyTurbine Technology Selection for Low-Wind Offshore Conditions in Mumbai High

The offshore waters of Mumbai High exceed 50 meters in depth, necessitating the use of floating wind turbines. The Mumbai High region presents unique aerodynamic conditions for wind development due to its moderate average wind speeds (6.5-7.2 m/s) and ageing offshore infrastructure. Selecting turbines tailored to low-wind-speed conditions is, therefore, critical to ensuring high energy yields, favourable capacity factors, and competitive levelized cost of energy (LCOE).

Comparative Analysis of Candidate Turbines. A detailed comparative matrix examined two turbine models: **Vestas V236-15.0 MW and ELEON 5 MW Turbine**. Each turbine was assessed across technical performance, economic feasibility, offshore deployment suitability, and environmental impact based on prior studies conducted in Maharashtra [25] and Malaysia [27].

Aspect	Vestas V236-15.0 MW	ELEON 5 MW
Document Source	Maharashtra Study [25]	Malaysian Study [27]
Selection Status	Selected	Selected

Technical Specifications:

Parameter	Vestas V236-15.0 MW	ELEON 5 MW
Rated Power	15.0 MW	5.0 MW
Rotor Diameter	236 m	155 m
Hub Height	Not specified	110 m
Tower Height	Not specified	90 m
Total Turbine Height	Not specified	187.5 m
Swept Area	43,742 m²	18,877 m²
Cut-in Speed	3.0 m/s	2.5 m/s
Rated Speed	Not specified	11 m/s
Cut-off Speed	31 m/s	21 m/s
Design Lifetime	30 years	25 years
Blade Configuration	115.5-meter blades	Not specified

Performance Metrics:

Metric	Vestas V236-15.0 MW	ELEON 5 MW
Annual Energy Production	Higher (due to larger swept area)	16.5/year
Capacity Factor	~45% (Maharashtra)	38% (at 6.8 m/s)
Wind Speed Environment	6.5-8 m/s (Maharashtra)	6.8 m/s (Malaysia)

Application & Deployment:

Category	Vestas V236-15.0 MW	ELEON 5 MW
Primary Application	Floating offshore wind farms	Hybrid power for O&G platforms
Foundation Type	Semi-submersible platforms	GICON®-TLP (Tension Leg Platform)
Water Depth	>50m (Maharashtra deep waters)	>50m (Malaysian waters)
Geographic Suitability	Deep waters, consistent winds	Low wind speed regions
Installation Approach	Large-scale wind farms	Single turbine on platform

Key Selection Rationales:

Turbine	Why Selected/Rejected
Vestas V236-15.0 MW	Selected: Highest capacity, largest swept area, proven offshore performance, reduces turbine count needed
ELEON 5 MW	Selected: Optimal for low wind speeds, suitable for platform mounting, proven TLP integration

Innovation & Technology Maturity:

Aspect	Vestas V236-15.0 MW	ELEON 5 MW
Technology Stage	Commercial/Advanced	Commercial/Specialized
Offshore Experience	Extensive	Platform-specific
Innovation Level	Cutting-edge large turbine	Low-wind specialization
Market Readiness	Ready for deployment	Ready for hybrid applications

Key Findings

Vestas V236-15.0 MW: The Vestas V236-15.0 MW emerged as the optimal choice for utility-scale projects in deeper waters (>50 m) off the western coast. With its massive swept area and high energy yield, this single turbine can achieve a capacity factor of 60% and is capable of producing 80 GWh per year, depending on site-specific conditions.[26] The recommended deployment strategy involves mounting these large-capacity turbines on semi-submersible floating platforms to maximise energy yield with minimal seabed disruption, aligning with modern offshore wind farm architectures.

ELEON 5 MW: ELEON 5 MW turbines proved particularly effective for brownfield integration on existing offshore platforms, especially in moderate wind conditions with 6.8 m/s average wind speed. These turbines achieve a 38% capacity factor when deployed using GICON®-TLP structures. [27] The optimal strategy involves retrofitting these turbines on existing Mumbai High platforms using tension leg platforms, which preserves capital by avoiding new foundation construction, accelerates deployment timelines, and leverages existing gas-turbine hybrid configurations already present on-site.

Subsea Transmission Infrastructure: Design and Routing Considerations for Mumbai High

The design and routing of subsea transmission infrastructure are central to ensuring reliable, safe, and cost-efficient integration of offshore wind energy into the existing oil and gas ecosystem of Mumbai High. Based on the Case 1 and Case 2 simulations from the study by Wei He et al. (2013), focusing on their design, objectives, and key technical insights. This can be directly used or adapted for the Mumbai High region.

Case 1: 20 MW Wind Farm. Configuration: Four 5 MW wind turbines operating alongside three 23 MW gas turbines on a single offshore platform.

Objectives:

- **Environmental Benefits:** Estimate long-term fuel savings and CO₂/NO_x emission reductions from wind power integration
- **Grid Stability:** Determine electrical grid stability when integrating four 5 MW wind generators
- **Integration Capacity:** Identify the maximum wind power capacity that can be safely integrated into the platform's stand-alone electrical grid

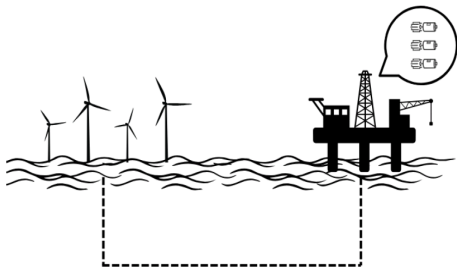


Figure 7—Case 1 Simulation

Simulation Results for Case 1 (20 MW Wind Farm)

This simulation revealed that integrating a 20 MW wind farm into the platform's electrical grid led to substantial environmental and operational benefits. Two strategies were assessed for managing the hybrid wind-gas turbine setup:

In this case, the simulation compared two operational strategies against a baseline with no wind power:

- **Equal Load Strategy:** All gas turbines operate continuously but at reduced load levels based on wind availability.
- **Start/Stop Strategy:** Gas turbines are cycled on and off depending on wind power availability.

Strategy	Wind Power	Fuel Savings	CO ₂ Reduction	NO _x Reduction
Baseline	0 MW	-	-	-
Equal Load	20 MW	18 Msm ³	39,720 tonnes	270 tonnes
Start/Stop	20 MW	24 Msm ³	53,790 tonnes	366 tonnes

Key Findings

- The **start/stop strategy** achieved the highest fuel savings and emission reductions—**40%**, compared to **30%** under the equal load approach.
- However, the start/stop operation resulted in the gas turbine starting and stopping **543 times per year** (~1.5 times daily), potentially **increasing mechanical wear** and **reducing operational lifespan**.
- Under both strategies, wind energy integration showed technical viability, provided backup systems were available to cover for wind intermittency.
- **Equal load strategy** offers a more conservative yet robust alternative for Mumbai High conditions, where operational risk minimisation is key. In this strategy:
 - All gas turbines continue to run at all times.
 - When wind is available, each gas turbine reduces its output in equal measure.
 - Remaining demand is proportionally distributed among the turbines.

These findings reinforce that a **20 MW modular wind integration** using an equal load control strategy is both technically and operationally suited for Mumbai High's offshore platform clusters. It balances performance with equipment longevity while enabling significant emissions reductions and fuel savings.

Case 2: 100 MW Wind Farm. Configuration: 20 Units of 5 MW wind turbines connected to five nearby offshore platforms via subsea power cables.

Objectives:

- **Platform-Specific Capacity:** Assess the maximum wind power that can be integrated into each platform
- **Multi-Platform Stability:** Electrical grid stability in terms of the control strategy and interconnecting grid topology.

Three Grid Topologies Test: The researchers tested three different ways to connect the five platforms:

- **"Star" Topology:** All platforms connect to a central hub - Not ideal for 100 MW wind farm due to limited fault tolerance.

- **"Star-F" Topology:** Modified star with additional connections - Best overall choice for a 100 MW wind farm — balances cost, complexity, and performance.
- **"Mesh" Topology:** Multiple interconnections between platforms - Technically best, but may be overengineered for 100 MW unless reliability is top priority.

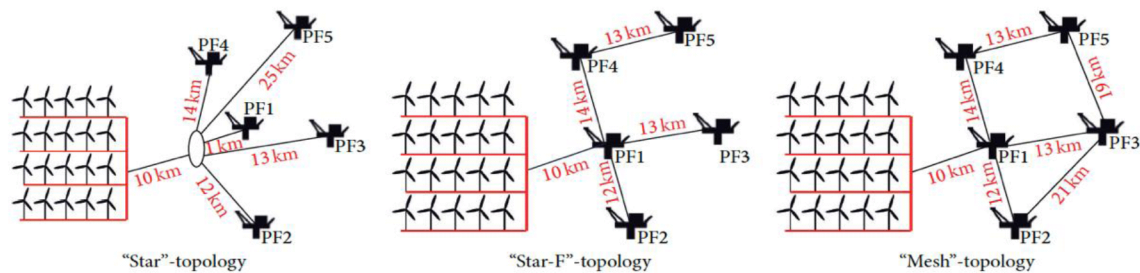


Figure 7a—Three different interconnections of the five platforms.

Table 1—Topology Comparison for Medium-Sized Wind Farms

Criteria	Star Topology	Star-F Topology	Mesh Topology
Grid stability	Moderate	Good	Excellent
Resilience to failure	Low	Moderate	High
System complexity	Low	Medium	High
Installation cost	Low	Medium	High
Scalability	Limited	Good	Excellent
Simulation performance (study)	Weakest	Acceptable	Best

Simulation Results for Case 2 (100 MW Wind Farm). **Platform Integration Limits:** The analysis determined maximum wind power integration limits for each platform by evaluating the critical "loss of all wind power" scenario. The system must maintain stability when wind generation suddenly disappears, establishing clear capacity thresholds for safe operation similar to the 20 MW Case 1 findings.

Optimal Grid Topology Selection: The Star-F Topology emerged as the most suitable choice for medium-sized (100 MW) offshore wind farms, offering an optimal balance between robustness, grid stability, cost-effectiveness, and system complexity. While Mesh Topology provides superior reliability for critical applications or harsh environments, the Star-F configuration delivers adequate performance at reasonable cost for most installations.

Complete wind power loss represented the most severe disturbance scenario, yet the system maintained stability across all three grid topologies.

Hybridisation Strategy: Synchronising Wind Energy with Gas Turbine Generators

The hybridisation strategy for Mumbai High offshore platforms has to draw inspiration from pioneering projects like Hywind Tampen, the world's first floating wind farm designed to power offshore oil and gas platforms. The core principle must involve creating a seamless integration between renewable wind energy and conventional gas turbine generators to ensure a continuous, reliable power supply while significantly reducing carbon emissions.

Wind-First Operation Framework. The proposed hybrid system follows a wind-priority operational model where offshore wind turbines are designated as the primary power source. Permanent Magnet Synchronous Generators (PMSG) and MPPT algorithms ensure maximum energy harvesting even under fluctuating wind conditions. [29]

Hybrid Power Management and Control Strategy. The hybrid system at Mumbai High follows a structured power generation hierarchy to ensure reliability and efficiency. The **primary energy source** is the floating offshore wind turbines equipped with **Permanent Magnet Synchronous Generators (PMSGs)**. When wind power is insufficient, **gas turbine generators** serve as the **secondary source**, providing backup and supplementary power. A **tertiary layer**, comprising **energy storage systems**, is planned for future integration to enhance system stability and dispatchability.

To maximise energy capture from the wind, the system employs the **Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT)** algorithm, adapted for offshore conditions. This control strategy dynamically adjusts the DC bus voltage to optimise power extraction under fluctuating wind speeds and sea states, ensuring stable and efficient operation. [29]

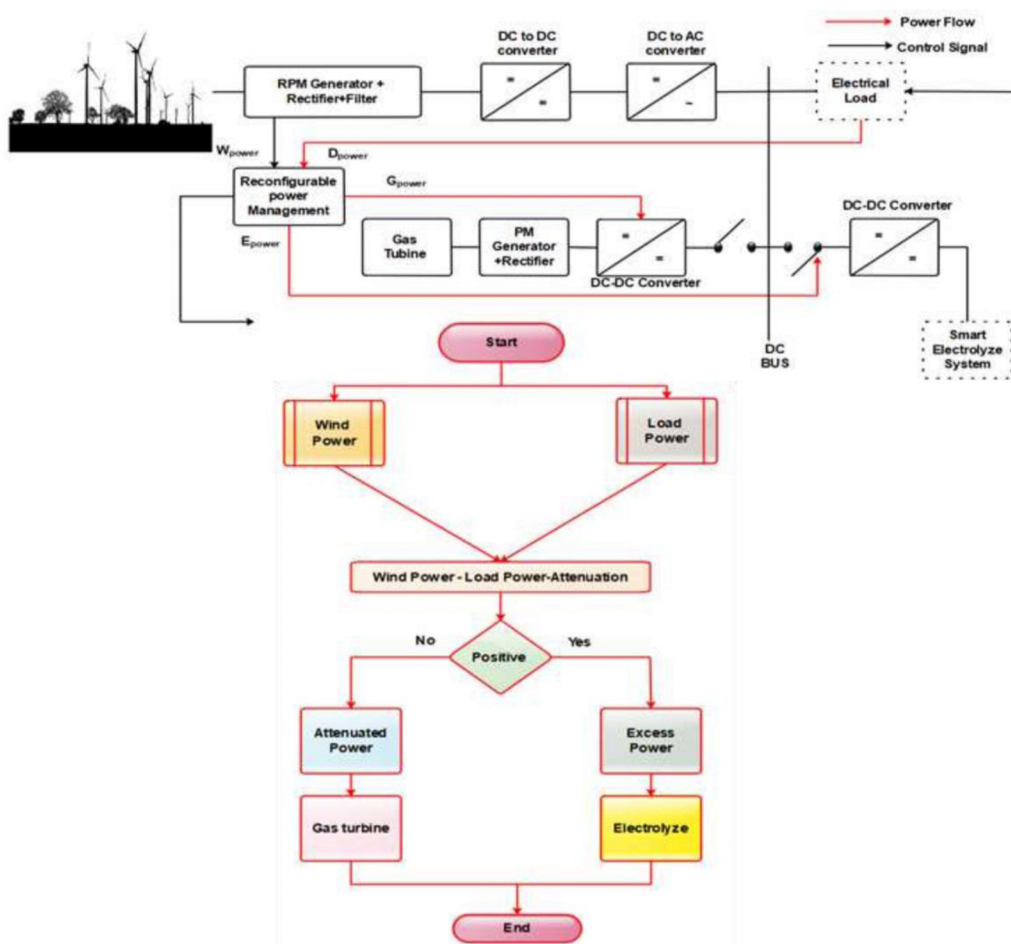


Figure 8—Proposed hybrid power management architecture [29]

Technologies for Offshore Power Transmission

The hybrid offshore system for Mumbai High requires robust and reliable power transmission technologies to efficiently transfer power across varying distances and environments. This section categorizes offshore transmission into two sub-types: short-distance (wind farms to platforms) and long-distance (platforms to shore), and discusses their technological components and challenges.

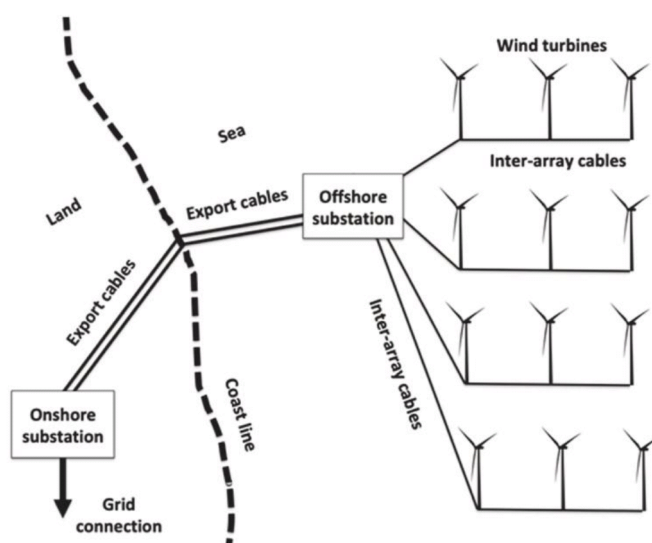


Figure 9—Shows the main transmission assets for a typical offshore wind farm.

Short-Distance Offshore Power Transmission: Wind Farm to Offshore Platforms. Intra-array or "array cables" are used to collect power from individual wind turbines and route it to a central offshore platform or substation. These cables typically operate at medium voltages (33 or 66 kV AC) and must withstand harsh subsea conditions. [31]

Technical Specifications:

- **Voltage Rating:** Array cables typically operate at 33 kV, though some developers are considering 66 kV systems for higher capacity turbines. [34]
- **Cable Construction:** Three-core AC submarine cables with XLPE insulation and protective armouring. [31]

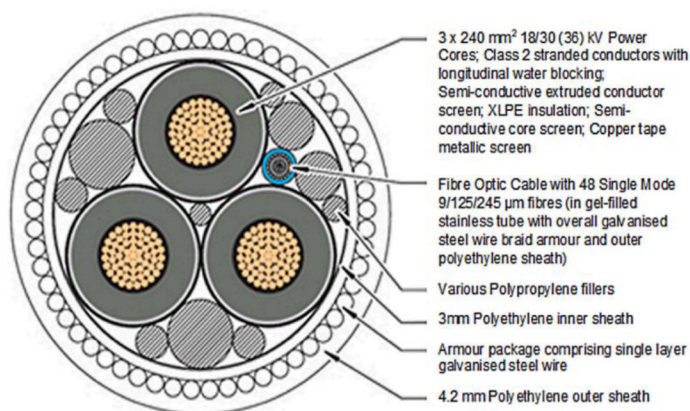


Figure 10—Schematic cross-section of an inter-array cable 33 kV [31]

Long Distance Offshore Power Transmission: Transmission of Power from Offshore to Shore. To transmit collected energy to the mainland grid, export cables are used—typically at higher voltages and sometimes over distances up to 100 km.

Technical Requirement:

- **Transmission Distance:** Up to > 150 km from shore for the Mumbai High location

- **Power Capacity:** Must handle combined output from multiple wind turbines plus existing platform power requirements

HVAC vs. HVDC:

- HVAC (High Voltage AC) cables suffer from high capacitive losses at sea due to reactive power; losses can reach up to 60% per 100 km for 750 kV HVAC sea cables. [32]
- HVDC (High Voltage DC) cables, by contrast, offer lower losses (~3% per 1000 km) and require less infrastructure—making them ideal for distances beyond 50 km offshore. [32]

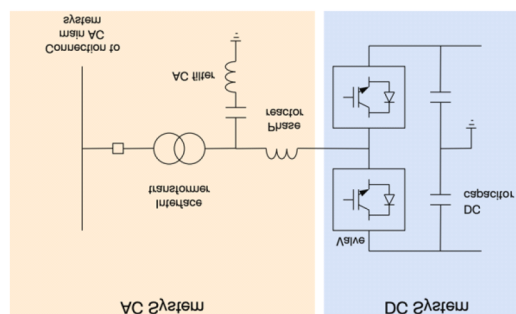


Figure 11—Example voltage source converter (VSC) [33]

Technological Choices:

- HVDC systems can use either Current Source Converters (CSCs) or Voltage Source Converters (VSCs). For Mumbai High, VSC technology may be preferred due to its smaller footprint and suitability for integration into hybrid microgrid systems. [33]

Three-phased Hybrid offshore implementation strategy for Mumbai High

A successful transition to hybrid offshore energy in the Mumbai High basin demands a structured deployment plan that mitigates operational risks while leveraging existing oil and gas assets. A **three-phased implementation strategy** is proposed to progressively scale renewable energy integration—from isolated platform-level pilots to a grid-connected offshore wind-gas hybrid system with HVDC shore export.

Phase I: Pilot Integration and Infrastructure Reuse

Objective: Establish a technically and economically feasible hybrid model by integrating wind power with the existing Gas Turbine Generators (GTGs) of the oil and gas platform. This phase is critical for demonstrating the technical feasibility, operational reliability, and environmental benefits of hybrid systems while leveraging current assets to minimise capital expenditure and project risk.

Key Elements:

- **Site Selection:** Choose 5 platforms of Mumbai High with high structural integrity and proximity to wind zones (6.5–8 m/s), such as those identified in *Feasible Region 1*.
- **Hybrid Architecture Setup:** Identify & Install the maximum wind power capacity (100 MW approx.) that can be safely integrated with the selected platforms and connected via AC array cables (66 kV) to the platform.
- **Operational Logic:** Implement an **Equal Load Strategy**, where GTGs run continuously at lower loads and proportionally adjust output based on wind availability—shown to reduce fuel consumption by ~30%.
- **Power Management System:** Implement basic MPPT-based wind control and real-time source coordination with wind-first logic and GTG fallback.

Expected Outcomes:

- Validate synchronisation strategy (Equal Load with wind-first logic).
- Demonstrate synchronisation, emissions reduction, and control stability.
- Identify O&M challenges unique to the Indian offshore environment (humidity, monsoons, salt spray).
- Quantify emissions reduction and economic savings (~30–40% fuel saving based on Case 1 simulation).
- Collect data for scaling in future phases.

Phase II: Smart Power Hub Control System Deployment

Objective: Create a distributed energy network with smart Power hub control systems and subsea transmission infrastructure.

The Smart Power Hub represents an advanced offshore electrical network paradigm designed to interconnect multiple oil and gas platforms and floating offshore wind farms into a cohesive, intelligent power system. Unlike traditional isolated power generation on individual platforms, the Smart Power Hub enables dynamic power sharing, improved grid stability, and optimised utilisation of renewable resources.

Key Activities:

- **Wind Farm Expansion:** Deploy 100 MW wind clusters around pilot platforms.
- **Advanced PMS Deployment:** Integrate full-fledged smart grid solutions with load forecasting, AI-assisted dispatch, and redundancy logic, based on successful models such as Hywind Tampen's control architecture. [35]
- **Subsea Cabling Strategy:** Connect turbine clusters with subsea array cables to offshore platforms; implement redundant routes to improve reliability. Preferably in Star–F Topology.
- **Begin HVDC Planning:** Start infrastructure and routing studies for long-distance shoreward power export.

Expected Outcomes:

- Optimised hybridisation at the cluster level.
- Reduction in platform-level blackout risks.
- Create data-backed models for GTG ramping wear and offshore wind variability.
- Greater emissions savings (up to 40%) with real-time hybrid flexibility.

Phase III: Shore Power Export

Objective: Establish a **grid-connected renewable power corridor** from offshore Mumbai High to the western coast shore via **HVDC transmission**, while integrating energy storage and smart hub functionalities.

Key Elements:

- **HVDC Deployment:** Commission offshore converter stations and lay ± 320 –500 kV HVDC export cables to the shore – suitable for long distances and high efficiency
- **Offshore Energy Hub Configuration:** Retrofit Power Hub to act as a **multi-energy control centre**, coordinating power from turbines, GTGs, and potential battery/hydrogen storage.

Expected Outcomes:

- 60% decarbonisation of Mumbai High operations.
- Offshore-to-onshore clean energy supply at the grid scale.

Economic Feasibility of Hybrid Offshore Wind System for Mumbai High

Introduction and Methodology

This section presents a comprehensive techno-economic analysis of a hybrid offshore wind system for the Mumbai High region, utilising established financial modelling methodologies for offshore renewable energy projects. The analysis employs the Levelized Cost of Energy (LCOE) framework and Net Present Value (NPV) assessment to evaluate project viability under Indian market conditions.

Reference Case and Scaling Methodology. The economic model is based on the Norwegian Sea floating wind farm study by Bjørni et al. (2023), which provides validated cost structures for floating offshore wind technology. A systematic scaling approach was employed to adapt the Norwegian reference case (50 turbines, 750 MW) to the Mumbai High configuration (7 turbines, 105 MW), yielding a scaling factor of 0.14.

Project Configuration and Technical Specifications.

System Architecture:

- **Total Capacity:** 105 MW
- **Turbine Configuration:** 7 × Vestas V236-15.0 MW floating wind turbines
- **Foundation Type:** Semi-submersible floating platforms
- **Water Depth:** 40-100 meters
- **Distance from Shore:** >150 km

Wind Resource Assessment:

- **Average Wind Speed:** 6.5-8.0 m/s at 100m hub height
- **Capacity Factor:** 45% (adjusted for wake losses and availability)
- **Annual Energy Production (AEP):** 414,630 MWh/year

Table 2—Turbine Technical Parameters

Parameter	Specification
Rated Power	15 MW
Rotor Diameter	236 m
Hub Height	100 m
Cut-in Speed	3 m/s
Rated Wind Speed	11.1 m/s
Cut-out Speed	31 m/s
Design Life	30 years

Capital Expenditure (CapEx) Analysis

Cost Structure and Scaling Methodology. The CapEx analysis employs a bottom-up cost estimation approach, with adjustments for Indian market conditions and water depth variations. The base Norwegian study costs were systematically scaled and adjusted using the following methodology:

Table 3—Base Cost Categories (Norwegian Reference - 50 Turbines)

Cost Component	Norwegian Cost (M \$)	Scaling Factor	Adjusted Cost (M\$)	Rationale
Electrical Grid	384.3	0.14	53.8	Linear scaling
Electrical Installation	221.4	0.14	31	Linear scaling
Turbines	759	0.14	106.3	Linear scaling
Substructure	1,275.70	0.1	127.6	Further reduction (depth advantage)
Substructure Installation	56.7	0.07	5.7	Further reduction (proximity advantage)
Mooring System	218.1	0.07	21.8	50% reduction (shallower water)
Soft Costs	653.4	0.14	91.5	Linear scaling
Project Development	203.9	0.14	28.5	Linear scaling
Subtotal	3,770.50	-	465.3	-

Regional Cost Adjustments for Mumbai High. Mumbai High benefits from lower water depths and existing infrastructure, reducing substructure and logistics costs. However, a 20% market premium is applied to account for import duties, regulatory costs, and currency risks. This brings the total **CapEx to \$558.4 million**, with a specific **CapEx of \$5.31 million/MW** and **\$79.8 million per turbine**.

Final CapEx Calculation:

Base Adjusted Cost: \$465.3 million

Market Premium (20%): \$93.1 million

Total CapEx = \$465.3 million + \$93.1 million = \$558.4 million

Operational Expenditure (OpEx) Analysis

The base OpEx of \$118,000/MW/year (Norwegian benchmark) is adjusted for Mumbai High with lower labour cost, lower vessel costs, and a 20% material cost premium, which is considered to be OpEx of **\$100,000/MW/year** or **\$10 million annually** for the 105 MW system.

Financing Structure and WACC Analysis

Weighted Average Cost of Capital (WACC). Given the higher risk profile of offshore renewable projects in emerging markets, the project's **WACC is estimated at 8.5%**, reflecting both equity and debt costs adjusted for sector and country risk.

Viability Gap Funding (VGF). To enhance bankability, the project utilises India's Viability Gap Funding (VGF) scheme under the Public-Private Partnership (PPP) model. It secures a primary VGF of **\$111.7 million (20%)**, along with an additional government support of **\$111.7 million (20%)**, bringing the total support to **\$223.4 million**, covering **40% of the total project cost**.

Fixed Charge Rate (FCR).

$$FCR = WACC / [1 - (1 + WACC)^{-n}] FCR = 9.305\%$$

Revenue Stream Analysis

Table 4—Platform-Wise Energy Demand and Fuel Consumption Profile

Platform Name	Total GT Available	Capacity in MW	Total Load of Platform in MW	Fuel Consumption BY GTs
Platform 1	7	75.6 MW	26.8 MW	3,05,252 SCMD
Platform 2	4	40 MW	11.125 MW	1,26,714 SCMD
Platform 3	4	35.8 MW	7.4 MW	84,287 SCMD
Platform 4	3	22.2 MW	5.2 MW	59,228 SCMD
Platform 5	7	50 MW	14.04 MW	1,59,915 SCMD

Primary Revenue: Avoided Fuel Costs. The project's primary revenue stream arises from savings in fuel costs by replacing a portion of gas turbine generation with offshore wind energy. Across five platforms, the total connected load is **64.6 MW, taking it (approx. 70 MW)**, resulting in an annual energy requirement of approximately 613,200 MWh. Assuming **50% of this demand is met by wind energy**, the displaced generation is **306,600 MWh/year**.

The associated gas consumption across all platforms is **735,396 SCMD**, and displacing 50% results in avoided consumption of **367,698 SCMD**. This equates to **12,979.7 MMBtu/day**, and at a gas price of **\$12.60/MMBtu**, the daily savings amount to **\$163,544.7**. Annually, this results in approximately **\$59 million** in avoided fuel costs.

Secondary Revenue: Carbon Credit Monetisation. With a grid emission factor of **0.934 tCO₂/MWh**, the annual emission reduction from 306,600 MWh of wind energy is **286,364 tonnes of CO₂**. At an average carbon credit price of **\$2.35/tCO₂** (Q1 2024), this yields an additional revenue of **\$0.7 million per year**.

Revenue Escalation. Fuel cost savings and carbon credit revenues are projected to escalate at 3% annually, taking into account gas price inflation, rising carbon credit values, and stricter emission regulations.

Financial Performance Analysis

LCOE Calculation.

$$LCOE = (Net\ Investment \times FCR + Annual\ OpEx) / Annual\ Energy\ Production$$

The Levelized Cost of Energy (LCOE) is estimated at **\$99.5/MWh**, based on an annual cost of **\$41.17 million** (including \$31.17 million in capital recovery and \$10 million in OpEx) and an Annual Energy Production (AEP) of 413,910 MWh.

Comprehensive Cash Flow Analysis. The financial analysis is based on a 30-year discounted cash flow (DCF) model. The base annual revenue is estimated at **\$59.7 million**, primarily from avoided fuel costs (**\$59.0 million**) and carbon credit sales (**\$0.7 million**), with revenues projected to escalate at **3% annually** to reflect expected increases in fuel prices and carbon credit values. On the cost side, the model assumes an initial investment of **\$335.0 million** (after accounting for Viability Gap Funding) and a constant annual operating expenditure of **\$10.0 million**. A discount rate of **8.5%**, reflecting the project's weighted average cost of capital (WACC), is applied in [Table 5](#) to evaluate the project's financial viability.

Table 5—Detailed Cash Flow Projections for selected years

Year	Revenue (M\$)	OpEx (M\$)	Net Cash Flow (M\$)	Present Value (M\$)	Cumulative PV (M\$)
0	0	0	-335	-335	-335
1	59.7	10	49.7	45.8	-289.2
5	67.2	10	57.2	38	-125.8
8	73.42	10	63.42	33.02	-21.89
9	75.63	10	65.63	31.49	9.6
10	77.9	10	67.9	30	39.6
15	90.3	10	80.3	23.6	170
20	104.7	10	94.7	18.5	272.3
25	121.4	10	111.4	14.5	352.4
30	140.7	10	130.7	11.3	415

Advanced Financial Metrics. The project demonstrates strong financial viability based on key investment metrics. The **Net Present Value (NPV)** is estimated at **\$415 million**, with an **Internal Rate of Return (IRR)** of **18%**. The **payback period** for the initial investment is projected to be **9 years**, indicating a robust return profile for investors.

Economic Viability Assessment Summary

The techno-economic analysis demonstrates that the proposed hybrid offshore wind system at Mumbai High is both financially and strategically sound. With a **Net Present Value (NPV) of \$415 million**, the project shows clear value creation, while an **Internal Rate of Return (IRR) of 18%** significantly exceeds the **cost of capital (8.5%)**, indicating strong investment potential. The **9-year payback period** is well within acceptable limits for infrastructure projects, and the **Levelized Cost of Energy (LCOE) of \$99.5/MWh** is competitive with conventional generation options.

Beyond the financial metrics, the project presents a compelling investment opportunity. It offers attractive returns, aligns closely with India's renewable energy targets, and serves as a technology demonstrator for future offshore wind developments. Additionally, it delivers environmental benefits that support corporate sustainability goals and contribute to energy security by diversifying the power supply mix.

Investment Recommendation: Based on its strong economic fundamentals and strategic relevance, it is recommended that the project advance to the detailed feasibility and engineering design phase.

Future Technology Cost Projections and Market Evolution

Floating Offshore Wind Cost Reduction Pathway (2025-2030)

The floating offshore wind industry is experiencing rapid technological maturation driven by economies of scale, supply chain optimisation, and technological innovation. Industry analysis by BVG Associates and Catapult projects significant cost reductions across both capital expenditure (CAPEX) and operational expenditure (OPEX) through 2030, fundamentally improving the economic viability of projects such as Mumbai High.

Based on comprehensive market analysis, floating offshore wind CAPEX is projected to decline from current levels of \$3,900-4,400/kW to \$2,400-2,900/kW by 2030, representing a 35-40% reduction. The primary cost reduction drivers include turbine technology advancement with larger, more efficient units achieving higher capacity factors, manufacturing scale economies from increased production volumes,

supply chain maturation through localised manufacturing and optimised logistics, standardised floating platform designs with materials optimisation, and improved installation efficiency through enhanced vessel availability and procedures.

Operational expenditure optimisation follows a similar trajectory, with current OPEX ranges of \$70-90/kw projected to decrease to \$55-75/kw by 2030, representing a 25-30% reduction. Key enablers include predictive maintenance systems utilising AI-driven condition monitoring, digitalisation through remote monitoring capabilities, specialised O&M vessel optimisation, and improved component reliability extending service intervals.

Mumbai High Project Cost Evolution Analysis

Applying industry cost reduction trends to the Mumbai High project configuration yields substantially improved economic projections for 2030 implementation scenarios. The revised analysis indicates total CAPEX reduction from \$558.4M to \$378.0M (32% reduction), annual OPEX decrease from \$100k/MW/year to \$65k/MW/year (35% reduction), and LCOE improvement from \$99.5/MWh to \$67.3/MWh (32% reduction), resulting in enhanced NPV of \$627.8M compared to the baseline \$415.0M.

Industry Learning Curve Application

The offshore wind industry demonstrates a consistent 13% learning rate, where costs decrease by 13% for every doubling of cumulative installed capacity. This learning curve presents significant strategic opportunities for sequential project development across India's offshore wind potential. The Mumbai High project serves as an anchor for broader offshore wind development, with cumulative benefits including technology transfer, reducing risks in subsequent projects, supply chain development, establishing local capabilities, operational excellence through proven O&M strategies, and financial optimisation, improving access to capital while reducing risk premiums.

The learning curve application positions Mumbai High as the foundation for portfolio development across Tamil Nadu (15-20 GW potential) and Gujarat (10-15 GW potential) offshore markets, with follow-on projects benefiting from 18-25% cost reductions relative to the initial Mumbai High deployment.

Conclusion

This study demonstrates the technical and economic feasibility of implementing a hybrid offshore wind system in Mumbai High, establishing a pathway for India's entry into the floating offshore wind market through strategic infrastructure reuse. Mumbai High's wind regime (6.5-8.0 m/s) supports commercial offshore wind development. The proposed 105 MW installation using Vestas V236-15.0 MW floating turbines achieves 45% capacity factor and 414,630 MWh annual generation. Integration with existing gas turbines through the Equal Load Strategy maintains grid stability while reducing fuel consumption by 30-40%. HVDC transmission technology provides scalable long-distance power export capability, with Star-F topology offering optimal balance between performance and cost for medium-scale deployments.

The project delivers strong financial returns with an NPV of \$415 million, IRR of 18%, and LCOE of \$99.5/MWh. The 9-year payback period, supported by India's Viability Gap Funding, creates an attractive investment opportunity. Primary revenue of \$59 million annually from avoided fuel costs provides immediate operational savings while achieving 286,364 tons of annual CO₂ reduction.

Infrastructure reuse reduces capital requirements by 32-45% compared to greenfield projects. Mumbai High's 100+ existing platforms provide established logistics networks and operational expertise, significantly reducing development risks. This model offers a globally applicable template for similar offshore oil and gas regions transitioning to renewable energy. The project establishes critical precedents for India's offshore wind sector, potentially catalysing 13% cost reductions for subsequent developments through learning curve effects. Integration enhances energy security through source diversification and supports India's Make in India initiative through local content requirements and technology transfer.

The three-phase implementation strategy provides a risk-mitigated pathway from 20 MW pilot demonstration to grid-scale deployment, with a modular approach allowing adaptive management based on early-phase learning.

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